

Liquidity Duration on Morpho Blue (Ethereum)

Capital Stickiness Across Address Types: A Duration Framework

Author: Ronith Yalamanchili

Date: 20 May 2026

Scope: Morpho Blue — Ethereum Mainnet Only

Data Sources: Dune Analytics (Query IDs: 6871840, 6872058, 6872073, 7502787, 7532526, 7535425, 7536232)

Coverage: 29 monthly depositor cohorts (Jan 2024 – May 2026), 24 loan tokens, 124 weeks of flow data, balance-based capital survival segmented by EOA / contract / known-vault address type.

*Independent exploratory analysis based entirely on publicly available on-chain data.
Not investment advice.*

Executive Summary

Morpho Blue's aggregate TVL and survival metrics are dominated by contract allocators, professional infrastructure that rotates capital programmatically. Reading these numbers without address-type segmentation produces systematically wrong conclusions about the protocol's depositor base.

The core finding: Morpho has a tiered capital stack. A stable, long-duration core of vault-mediated LPs and direct EOAs is obscured behind a high-velocity allocator layer that dominates dollar inflow during growth waves. The two layers have fundamentally different duration profiles and must be evaluated separately.

Key Findings

1. Aggregate survival rates are allocator artifacts, not user flight. For the three largest cohorts (Sep–Nov 2025, combined \$789M+ gross inflow), the `all` 30-day survival collapsed to <2%. This is entirely a contract-bucket story: allocators held 94–97% of dollar weight and rotated to zero. Direct EOA and vault LP retention in the same cohorts reached 34.3% and 41.7% respectively. The rollup does not describe end-user behavior.

2. EOA capital survival is bimodal, not a single parameter. Across 17 cohorts with $\geq \$1\text{M}$ in direct EOA notional, 30-day survival splits into two clusters with a clean gap at $\sim 7.4\%$: 9 cohorts above 13% (typically 20–35%+) and 8 cohorts below 7% (sometimes <1%). Reporting a protocol-wide average masks this structural split entirely.

3. Vault LP stickiness is structurally superior after the contamination fix. Tracing ERC-4626 vault share transfers to true underlying EOAs (`eoa_underlying`) yields a median 30-day capital survival of 37–40%, materially above the contract bucket. A contamination audit (Dune query 7535425) found that 15.28% of raw `eoa_underlying` capital notional routed to contract recipients, not true EOAs. The EOA filter (query 7536232) removes this; figures in this report are EOA-only.

4. Vintage timing dominates curator brand in determining LP duration. Flagship (45.9% passive 90d, 155-day median hold) and Re7 Labs (35.8% passive 90d, 111 days) outperform Smokehouse (20.2%, 59 days), confirming curator-level effects are real. However, within a single curator (e.g., Gauntlet USDC Core), early 2024 vintages retained 50% at 90 days, while the Nov 2025 growth-wave cohort collapsed to 10% with a 6.8-day median hold. **The relevant unit of analysis is vault × curator × cohort vintage**, not curator alone.

5. The October 2024 regime break marks a structural shift in vault-LP behavior. Pre-Oct 2024 `eoa_underlying` 90-day survival averaged ~49% across cohorts. Post-Oct 2024, it fell to 11–19% and has not recovered. This coincides with the Morpho Vault ecosystem scaling and the entry of institutional allocators deploying capital at scale on a quarterly rotation cadence.

Metric Hierarchy

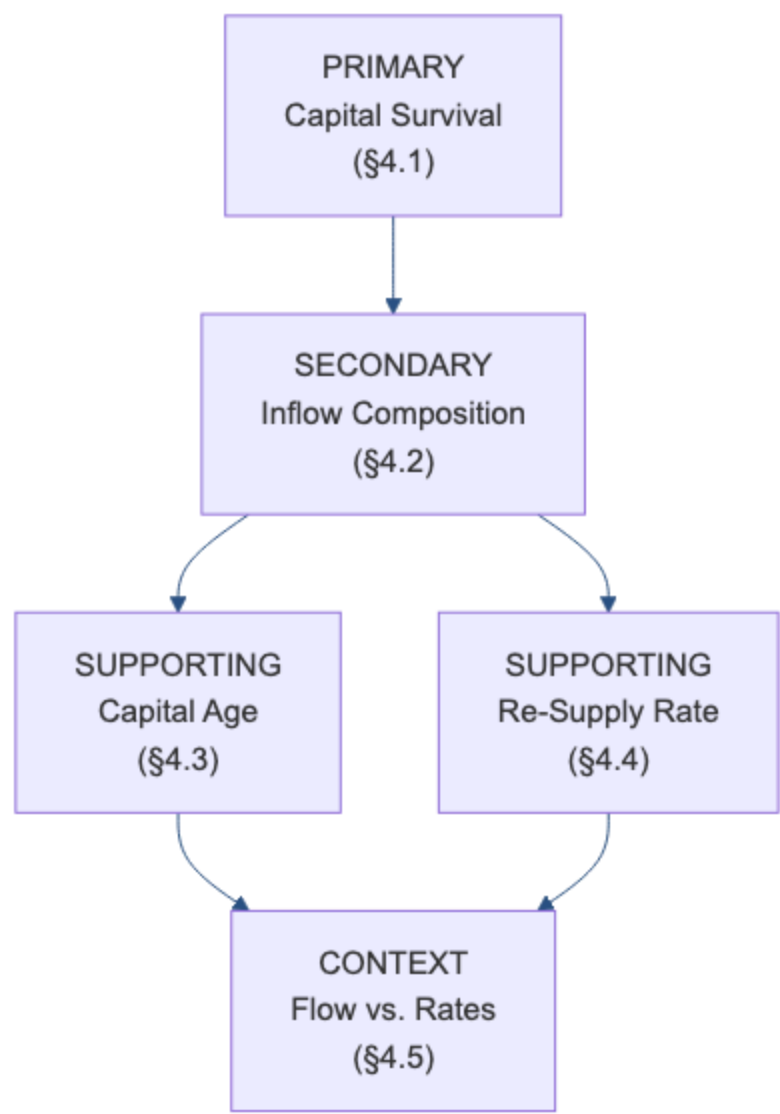
Tier	Metric	Question
Primary	Balance-based capital survival, by address type	Did deposited dollars remain? Split by who deposited.
Secondary	USD-weighted capital age (live-balance vs. historical-event)	How old is the average surviving dollar?
Secondary / behavioral	Cohort re-supply rate (30d / 90d)	Did depositors execute another supply within 30 or 90 days? (<i>Penalizes passive holders; not a stickiness proxy.</i>)
Tertiary / context	Flow sensitivity to rates	Did inflows persist during rate declines?

Address Taxonomy (Precedence: `eoa_underlying` > `known_vault` > `contract` > `eoa`)

Bucket	Definition
<code>eoa_underlying</code>	True underlying EOAs of Morpho vaults, resolved via vault share transfer tracking. Contract recipients excluded via <code>ethereum.creation_traces</code> .
<code>known_vault</code>	Deployed Morpho Vault acting as direct Morpho Blue market supplier. Dominant in 2024 H1; disappeared post-Oct 2024.

Bucket	Definition
contract	Smart contracts routing capital (allocators, routers, smart wallets). Dominant in 2025+.
eoas	Externally Owned Accounts supplying directly to Morpho Blue markets.
all	Dollar-weighted aggregate rollup across all buckets.

Figure 1 — Metric Hierarchy & Reading Order



2. Data & Methodology

This analysis tracks ERC-4626 vault LP activity on Morpho Blue (Ethereum Mainnet) using on-chain supply data from Dune Analytics. We trace vault share transfers to identify true underlying EOAs, apply a contamination filter to exclude contract recipients, and segment surviving capital by address type (`eoas` , `contracts` , `known_vaults` , `eoas_underlying`). The primary metric is balance-based capital survival: for each first-deposit cohort, we measure the fraction of initial USDC notional

still on deposit at 7, 30, 90, and 180 elapsed days. Secondary metrics include USD-weighted capital age (historical-event and live-balance approximations), cohort re-supply rates as an activity proxy, and weekly flow sensitivity to borrow rate changes. All survival figures in this report use the EOA-filtered `eoa_underlying` bucket unless noted otherwise. Detailed address taxonomy, cross-protocol calibration methodology, and the ERC-4626 vault share pipeline are described in Appendix A.

3. Findings

Dataset scope

Dataset	Count
Survival cohorts	29
Survival address types	<code>all</code> , <code>contract</code> , <code>eoa</code> , <code>eoa_underlying</code> , <code>known_vault</code>
Survival rows	440
Tokens (age proxy)	24
Flow weeks	124

3.1 Balance-Based Capital Survival by Address Type (*Primary Metric*)

Takeaway: The `all` rollup is dollar-weighted and tracks the dominant bucket. For large 2025 cohorts, that means contract allocators. EOA and `eoa_underlying` survival diverges materially from the headline in those cohorts.

Definition. For each first-deposit cohort, measure the share of initial notional outstanding at 7, 30, 90, and 180 elapsed days, by address bucket. Per-wallet net USDC balance is reconstructed from `lending.supply` events.

Table 3.1a — Capital Survival Rate: `all` Rollup (cohorts $\geq$ \$1M initial USDC)

Cohort	Initial USDC	7d	30d	90d	180d
2024-05	\$1.3M	96.87%	21.88%	21.88%	16.24%
2024-10	\$1.4M	0.00%	59.74%	0.00%	59.74%
2024-12	\$9.4M	71.70%	9.32%	1.21%	0.00%
2025-01	\$50.4M	11.23%	4.86%	4.39%	3.74%
2025-02	\$18.1M	65.65%	35.39%	20.91%	15.09%
2025-03	\$12.2M	43.96%	24.61%	15.26%	12.60%
2025-04	\$6.9M	24.85%	16.15%	19.43%	0.02%

Cohort	Initial USDC	7d	30d	90d	180d
2025-05	\$4.2M	61.75%	0.86%	0.03%	0.01%
2025-06	\$1.1M	7.54%	15.97%	1.76%	1.77%
2025-07	\$3.0M	82.72%	44.97%	21.92%	36.57%
2025-08	\$6.1M	67.66%	49.83%	32.46%	1.95%
2025-09	\$82.9M	1.35%	1.05%	0.01%	0.01%
2025-10	\$292.5M	1.83%	1.10%	0.22%	0.30%
2025-11	\$414.5M	1.77%	0.35%	0.40%	2.08%
2025-12	\$6.6M	27.60%	25.87%	4.25%	n/a
2026-01	\$5.3M	59.66%	42.17%	37.67%	n/a
2026-02	\$20.3M	86.53%	56.24%	19.22%	n/a
2026-03	\$9.1M	58.36%	39.52%	n/a	n/a

The <2% rollup for Sep–Nov 2025 is almost entirely a contract-bucket number. Table 3.1b disaggregates it.

Table 3.1b — 30-Day Capital Survival by Address Bucket

Cohort	Initial USDC	Contract \$ Share	All 30d	EOA 30d	Contract 30d	EOA Underlying 30d
2024-05	\$1.3M	78%	21.88%	n/a	0.00%	75.19%
2024-10	\$1.4M	60%	59.74%	0.00%	100.00%	44.60%
2024-12	\$9.4M	58%	9.32%	22.11%	0.00%	37.06%
2025-01	\$50.4M	2%	4.86%	4.84%	6.29%	31.07%
2025-02	\$18.1M	0%	35.39%	35.42%	15.22%	33.47%
2025-03	\$12.2M	25%	24.61%	7.00%	76.17%	24.56%
2025-04	\$6.9M	0%	16.15%	16.14%	100.00%	48.31%
2025-05	\$4.2M	0%	0.86%	0.83%	11.03%	43.88%
2025-06	\$1.1M	6%	15.97%	15.18%	27.43%	78.36%
2025-07	\$3.0M	37%	44.97%	61.44%	16.82%	44.30%
2025-08	\$6.1M	50%	49.83%	0.03%	99.83%	41.49%
2025-09	\$82.9M	97%	1.05%	34.29%	0.01%	41.74%
2025-10	\$292.5M	97%	1.10%	1.72%	1.09%	14.89%
2025-11	\$414.5M	95%	0.35%	6.59%	0.01%	33.14%

Cohort	Initial USDC	Contract \$ Share	All 30d	EOA 30d	Contract 30d	EOA Underlying 30d
2025-12	\$6.6M	7%	25.87%	20.04%	100.00%	35.30%
2026-01	\$5.3M	21%	42.17%	26.44%	100.00%	73.88%
2026-02	\$20.3M	86%	56.24%	9.89%	63.97%	33.29%
2026-03	\$9.1M	17%	39.52%	35.92%	57.62%	20.44%
2026-04	\$61.7M	90%	7.61%	8.56%	4.10%	8.90%

Pattern: When contract dollar share exceeds 90% (Sep/Oct/Nov 2025), the `all` rollup tracks contract behavior almost exactly. EOA and `eoa_underlying` diverge materially in 2 of 3 cases. The exception (Oct 2025) reflects market-wide de-risking where exit behavior was correlated across all address types.

Table 3.1c — Capital Half-Life by Address Bucket (first horizon at which survival < 50%)

Cohort	All	EOA	Contract	Known Vault	EOA Underlying
2024-05	≤ 30d	n/a	≤ 30d	> 180d	> 180d
2024-10	≤ 7d	≤ 7d	≤ 7d	> 180d	≤ 30d
2024-12	≤ 30d	≤ 7d	≤ 30d	> 180d	≤ 30d
2025-01	≤ 7d	≤ 7d	≤ 7d	n/a	≤ 30d
2025-02	≤ 30d	≤ 30d	≤ 30d	n/a	≤ 30d
2025-09	≤ 7d	≤ 7d	≤ 7d	n/a	≤ 30d
2025-10	≤ 7d	≤ 7d	≤ 7d	n/a	≤ 7d
2025-11	≤ 7d	≤ 7d	≤ 7d	n/a	≤ 7d
2026-01	≤ 30d	≤ 7d	> 90d	n/a	≤ 90d

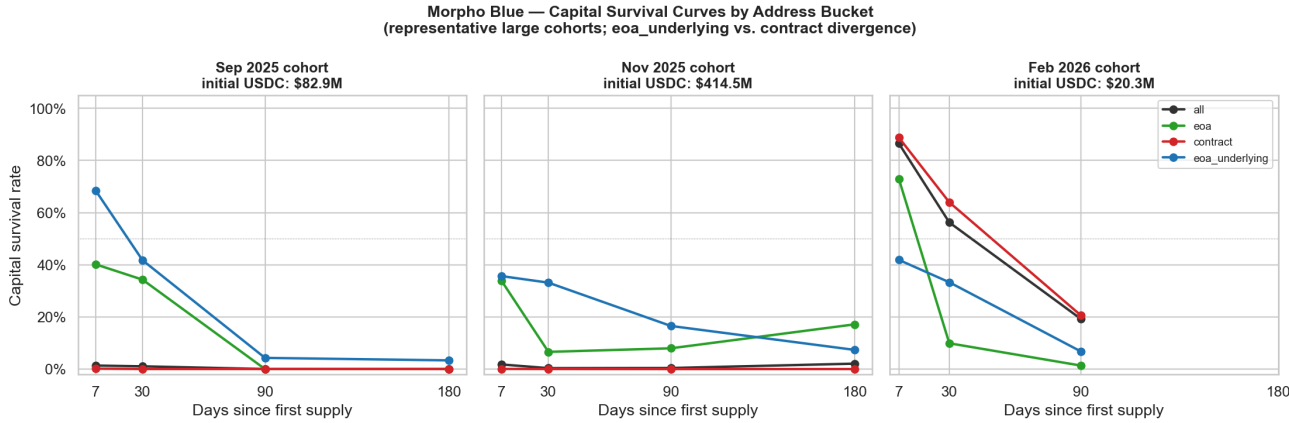
A ≤ 7d half-life means >50% of initial capital left within the first week.

`eoa_underlying` predominantly shows ≤ 30d half-life, materially longer than the contract bucket in most cohorts.

Figure 2 — Capital Survival Curves by Address Bucket (Sep 2025, Nov 2025, Feb 2026)

The September 2025 cohort (97% contract weight, \$82.9M inflow) shows the starkest split: contract 30-day survival at 0.01% versus `eoa_underlying` at 41.7%. The November 2025 cohort repeats the pattern at larger scale: 95% contract weight on \$414.5M in gross inflow, with contract survival again near zero while `eoa_underlying` held at 33.1%. February 2026 presents a narrower gap, with contract survival at 64.0% and `eoa_underlying` at 33.3%, reflecting lower allocator concentration in a smaller cohort. Across all three snapshots, `eoa_underlying` lines hold

materially above `contract` lines, confirming the structural separation in holding behavior between the two address buckets.

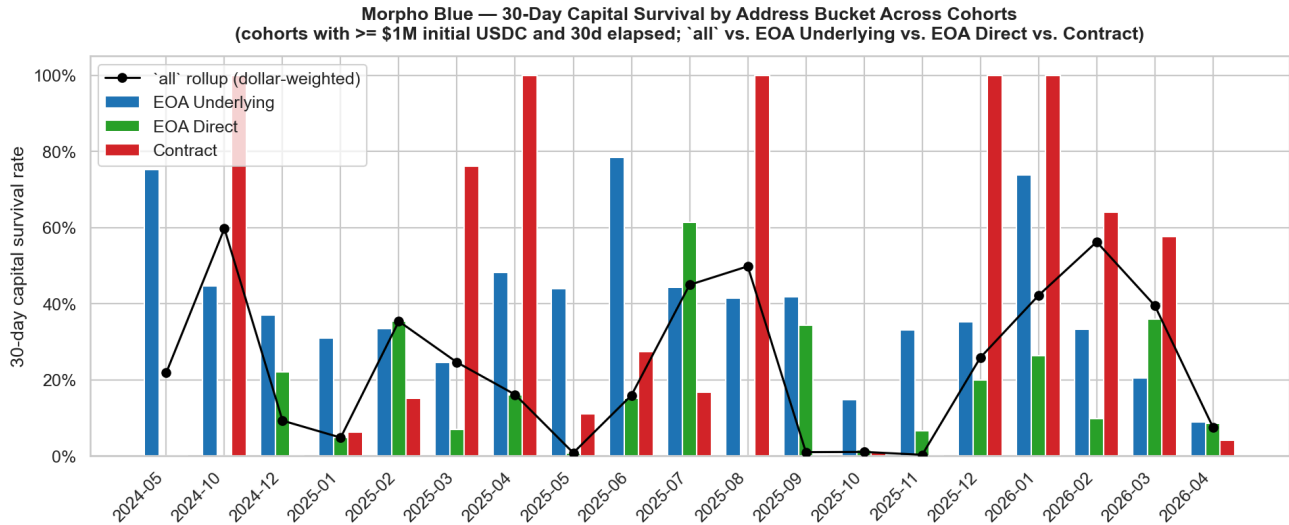


Morpho Blue — Capital Survival Curves by Address Bucket for representative large cohorts. Note the `eoa_underlying` vs. `contract` divergence.

3.1.1 TVL Rollups Are a Category Error for Duration Analysis

Takeaway: The aggregate `all` rollup is dollar-weighted and tracks whoever holds the most dollars. In large 2025 cohorts, that is always the `contract` allocator bucket.

Figure 3 — 30-Day Capital Survival by Address Bucket Across All Cohorts



Morpho Blue — 30-Day Capital Survival by Address Bucket Across Cohorts (cohorts $\geq \$1M$ initial USDC with 30d elapsed).

For Sep–Nov 2025 (combined \$789M+ inflow), the `contract` bucket held 95–97% of notional and rotated to near-zero within 30 days. Because the rollup is dollar-weighted, it mirrors this exactly: $<2\%$ at 30 days. In the same cohorts, direct EOA retained 34.3% (Sep 2025) and 6.6% (Nov 2025); `eoa_underlying` retained 41.7% (Sep 2025) and 33.1% (Nov 2025).

Contract rotation is programmatic risk management by professional allocators, not capital flight. Morpho's modular design explicitly enables curators to reallocate capital dynamically across markets; a 30-day near-zero contract survival rate means the system is working as designed.

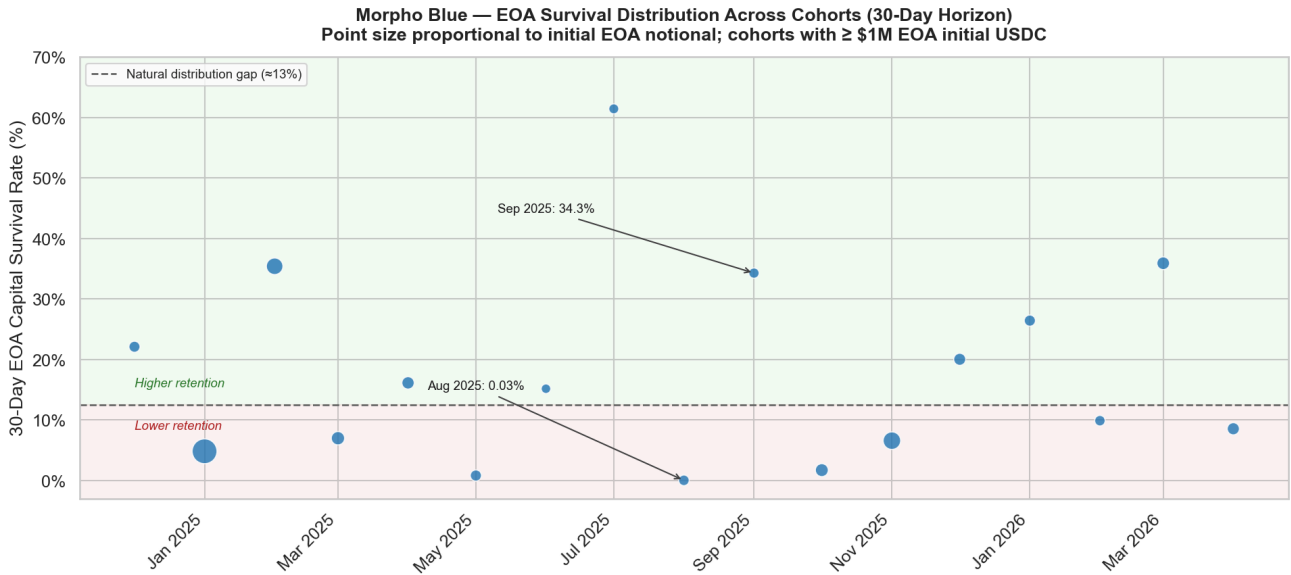
Caveat (Oct 2025 convergence): In October 2025 (\$292.5M cohort), the EOA bucket's 30-day survival also collapsed to 1.72%, converging with the contract bucket's 1.09%. This reflects market-wide correlated de-risking. The EOA-over-contract divergence is cohort-conditional and should not be assumed to hold universally.

3.1.2 EOA Capital Survival is Bimodal: No Single Stickiness Number

Figure 4 — EOA 30-Day Survival Distribution Across Cohorts (Bimodal Split)

Takeaway: 30-day EOA capital survival across 17 cohorts with ≥\$1M in direct EOA deposits splits into two clean clusters with a gap at ~7.4%. Reporting a protocol-wide EOA average hides the structural split.

- **Low-retention cluster (8 cohorts):** 30d survival <7% (including 0.03% in 2025-08, 1.72% in 2025-10)
- **High-retention cluster (9 cohorts):** 30d survival 20–35%+ (35.42% in 2025-02, 34.29% in 2025-09)



Scatter plot of each cohort's EOA 30-day capital survival (y-axis) vs. initial EOA notional (x-axis). Point size reflects cohort scale. Note the clean gap between clusters with no smooth continuity.

Cohorts split into two distinct groups: high-retention (20–35% at 30 days) and low-retention (<7%), with no cohorts occupying the middle range. A protocol-wide EOA average would land inside that empty gap and describe no individual cohort accurately: which cluster a cohort falls into appears driven by market conditions and curator composition at the time of entry, not by cohort size.

The bimodal pattern is inconsistent with either a "sticky" or "mercenary" single characterization of EOA depositors. The driver is not directly identified in this dataset; plausible candidates include macro/rate-regime timing at cohort entry, curator composition differences across months, or a liquidity event effect where large TVL inflows during growth waves attracted shorter-duration participants.

Wallet–capital survival gap (vault LPs, Jan 2024 – Jun 2025 mature cohorts):

Horizon	Gap (wallet% – capital%)
7d	–1 pp (inverted)
30d	+4 pp
90d	+3 pp
180d	+6 pp

A near-zero or slightly negative gap at short horizons indicates that Morpho's exits are clean: depositors who choose to withdraw do so fully, rather than leaving residual dust in their accounts. Over longer horizons, the gap widens to +6 percentage points, suggesting a two-population structure: a tail of small depositors who leave fractional amounts behind, alongside institutional allocators who manage positions deliberately and exit cleanly. This contrasts sharply with legacy lending venues where habituated users and complex contract approvals create wide, persistent positive gaps (+10 to +14 pp) across all horizons.

The bimodal distribution and near-zero wallet-capital gap together confirm a sophisticated, institutional-skewed user base in which direct EOA retention is cohort-dependent rather than homogeneous. Plausible candidate explanations include macro/rate-regime timing at cohort entry, curator composition differences across months, and inflow channel mix (direct EOA vs. vault-mediated).

3.1.3 The October 2024 Composition Shift

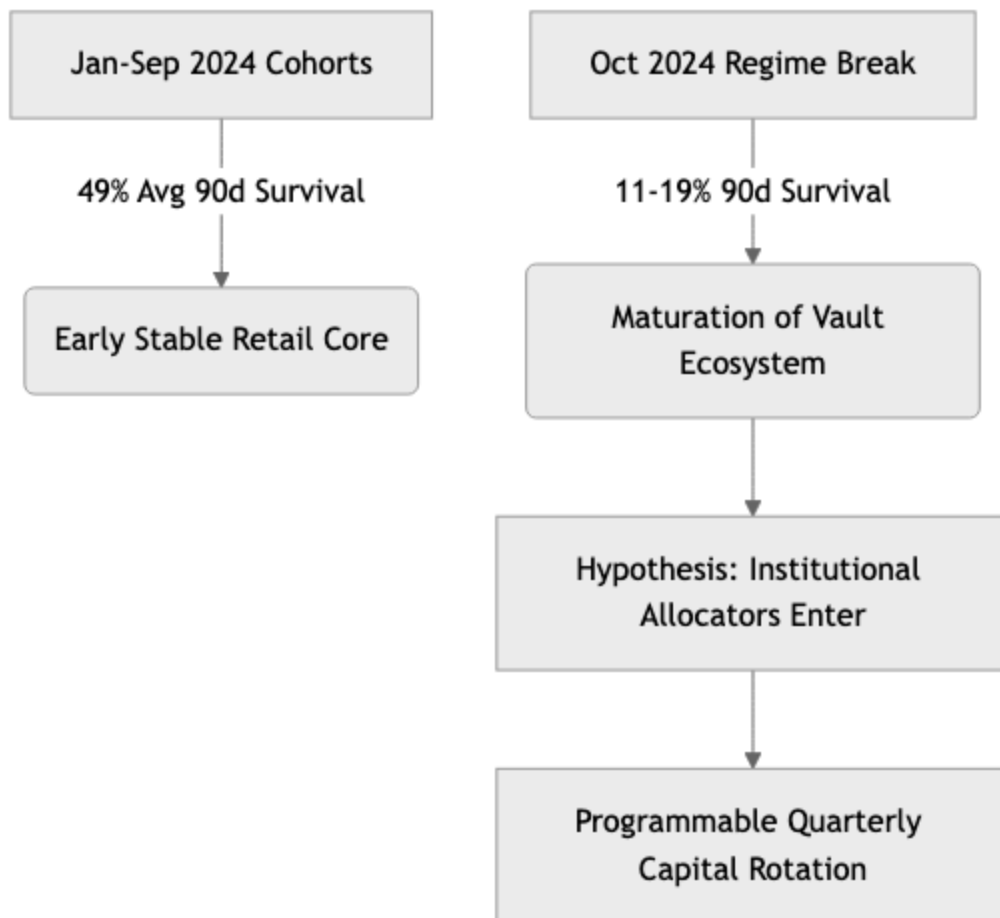
October 2024 coincided with Morpho Vault's transition from a small-scale deployment to a protocol with meaningful institutional allocator participation. New curators launched vaults with larger capacity targets, and institutional capital began flowing in at scale for the first time.

The 90-day eoa_underlying survival rate stepped down by ~38 percentage points in October 2024 and has not recovered.

Pre-Oct 2024 cohorts (Jan–Sep 2024): 90-day eoa_underlying survival ranged 22–72%, capital-weighted average ~49%.

Post-Oct 2024: collapsed to 11.4% (Oct 2024) and has remained in the 11–19% range for mature cohorts (e.g., 12.0% Jan 2025, 17.0% Feb 2025).

Figure 5 — October 2024 Composition Shift — Vault LP Survival Step-Change



Hypothesis: October 2024 marks the Morpho Vault ecosystem scaling from retail-oriented to institutionally-scaled deployment. The step-change in 90-day duration likely reflects the entry of institutional allocators deploying capital at scale and rotating on quarterly cadences. This is consistent with the routing architecture transition observed in Table 3.2. Vault-mediated LPs remain structurally stickier than direct contract allocators, but their average duration is no longer governed by early-adopter hold behavior.

3.2 Supply Inflow Composition: The `known_vault` → `contract` Transition

Takeaway: Morpho's routing architecture changed structurally at October 2024. Early cohorts were served by `known_vault` entries (vault contracts supplying directly). Post-Oct 2024, the `PublicAllocator` and `Bundler` routing infrastructure became dominant, and `known_vault` capital fell to zero as Morpho Vault moved one hop upstream. This creates a comparability problem across vintages.

Vault blindspot and contamination. When users supply via Morpho Vaults, the vault contract is recorded as the `on_behalf_of` address in Morpho Blue, causing direct on-chain analysis to misclassify all vault-mediated deposits as `contract`. Tracking ERC-4626 vault share transfers directly resolves this. The contamination audit (Dune query 7535425) found 15.28% of raw

`eoa_underlying` notional routed to contract recipients; the EOA filter in query 7536232 removes them.

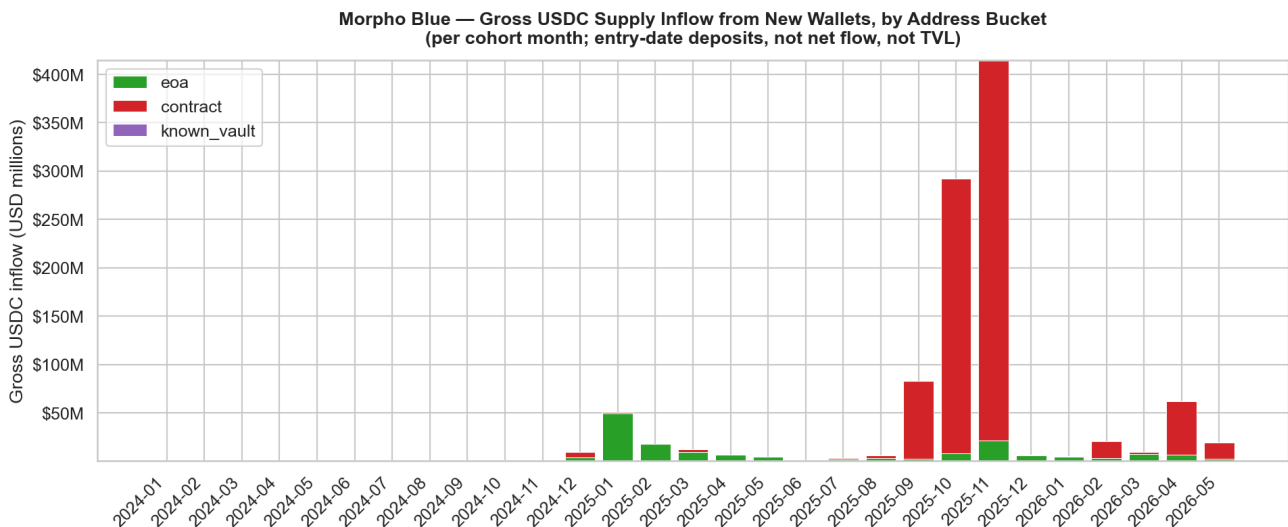
Median 30-day survival for EOA-filtered vault LPs: 37–40%. This is structurally superior to the contract allocator bucket, which frequently rotates to near-zero in the same timeframe.

Table 3.2 — Routing Architecture Transition (known_vault vs. contract % of entry capital)

Cohort	Total Inflow	known_vault %	contract %	Dominant Layer
2024-04	\$155.9K	0.1%	0.0%	EOA-dominant
2024-05	\$1.28M	21.9%	78.1%	Hybrid
2024-10	\$1.36M	0.0%	59.7%	Contract (allocator)
2024-12	\$9.37M	0.0%	57.9%	Contract (allocator)
2025-01	\$50.4M	0.0%	1.7%	EOA-dominant (surge)
2025-03	\$12.2M	0.0%	25.5%	Mixed
2025-09	\$82.9M	0.0%	97.0%	Fully allocator-routed
2025-10	\$292.5M	0.0%	97.2%	Fully allocator-routed
2025-11	\$414.5M	0.0%	94.9%	Fully allocator-routed

`known_vault` entries are absent from all cohorts after September 2024.

Figure 6 — Gross USDC Supply Inflow from New Wallets by Address Bucket



Morpho Blue — Gross USDC inflow from first-deposit cohorts, stacked by address bucket. The Sep–Nov 2025 contract (red) surge and near-absence of `known_vault` (purple) post-Oct 2024 are clearly visible.

Implication for cross-vintage comparisons: Early 2024 headline survival rates are inflated by vault-contract accounting (vaults don't withdraw unless a curator reallocates). 2025+ rates reflect allocator rotation. EOA-isolated figures serve as the cleanest cross-vintage comparable.

3.3 Curator × Vintage: Duration Is Vintage-Dominant

Takeaway: Curator identity is a real but secondary factor in LP duration. Within-curator vintage timing effects are large enough to overwhelm curator-level design differences.

Table 3.3 — Vault LP Share Activity by Curator

Curator	Vaults	LPs	Passive 90d	Passive 180d	Ever Redeemed	Median Hold
Flagship	3	772	45.9%	32.6%	82.8%	155d
Re7 Labs	6	1,247	35.8%	21.7%	88.8%	111d
Steakhouse Financial	5	3,694	28.4%	19.6%	74.4%	95d
Gauntlet	11	5,528	26.6%	17.5%	82.5%	92d
Usual	1	6,790	25.6%	14.2%	78.7%	80d
MEV Capital	4	1,287	24.9%	14.8%	82.8%	63d
Other / Unknown	7	1,845	21.9%	12.6%	72.2%	70d
Smokehouse	2	1,774	20.2%	12.7%	88.2%	59d

The curator spread is real: Flagship (45.9% passive 90d, 155d hold) vs. Smokehouse (20.2%, 59d). Institutional curators (Re7, Steakhouse, Gauntlet) cluster in the 26–36% range with 90–111 day medians.

But vintage timing dominates within curators:

Gauntlet USDC Core: Aug/Sep 2024 vintages → 50% passive 90d, 96–82 day median holds. Nov 2025 growth vintage → same vault, same curator → 10% passive 90d, 6.8-day median hold.

The Nov 2025 growth wave doubled the vault's LP count but brought in a cohort that rotated within days. Macro growth waves overwhelm curator-level design choices.

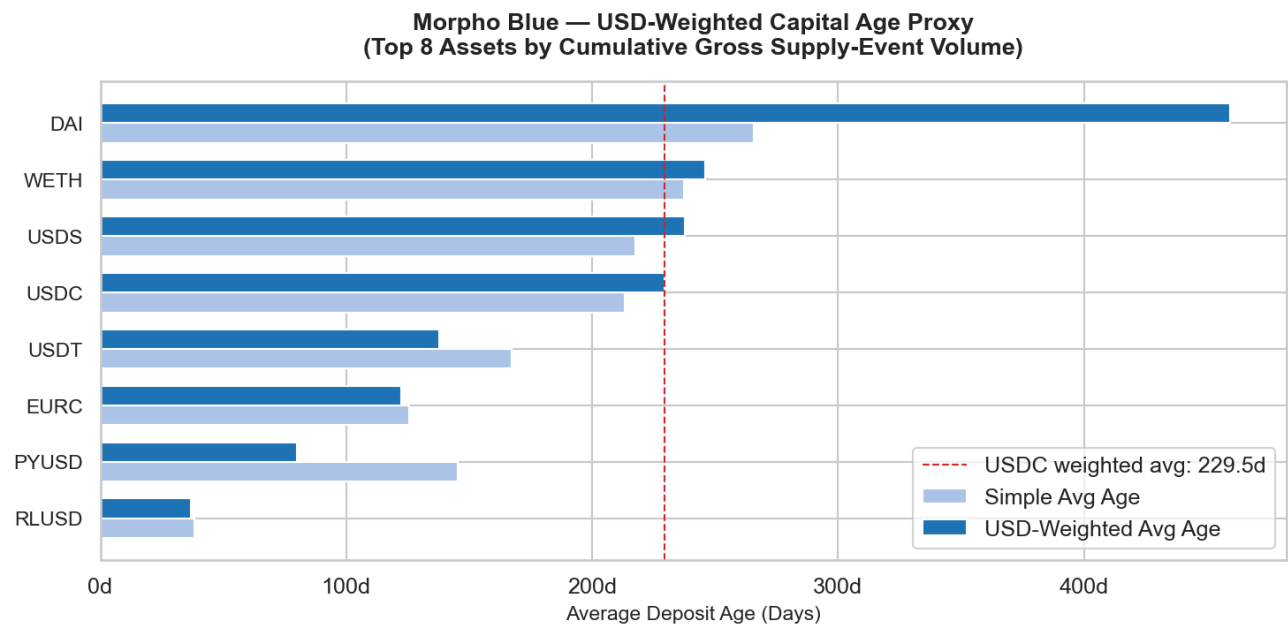
For partners evaluating credit duration: The relevant unit is **vault × curator × cohort vintage**. A single curator-level average is misleading when that curator's LP composition has shifted across vintages.

3.4 USD-Weighted Capital Age: Historical vs. Live-Balance

Takeaway: The historical-event-weighted age (229.5 days on \$239.2B gross USDC inflow) overstates live stickiness by ~93 days. Withdrawn deposits continue aging on paper. The live-

balance approximation (~137 days) is the operationally relevant figure.

Figure 7 — USD-Weighted Capital Age by Asset (Top 8)



Dashed red line marks the USDC USD-weighted average.

USDC Capital Age Summary

Metric	Value
Gross cumulative supply-event volume	\$239.2B
USD-weighted average age (historical-event)	229.5 days
Live-balance-weighted age (approximation)	136.7 days
Gap (historical – live)	92.8 days
Surviving USDC at latest elapsed horizon	\$40.7M
Surviving / gross-volume ratio	0.02%

Table 3.4 — Capital Age: Historical-Event vs. Live-Balance

Metric	Value
Live-balance-weighted age (approx.)	136.7 days
Historical-event-weighted age	229.5 days
Gap	92.8 days

The 92.8-day gap is telling: a substantial fraction of the cumulative supply-event series corresponds to deposits that have since been withdrawn but continue to age on paper. Even after the live-balance correction, the average surviving USDC dollar is ~4.5 months old, consistent with a meaningful duration tail but not the 7.5-month figure the historical-event proxy suggests.

Methodology note: The live-balance figure is a cohort-level approximation using the latest elapsed-horizon survival rate per cohort. Values in the 110–160 day range are plausible. Per-event reconstruction would refine precision.

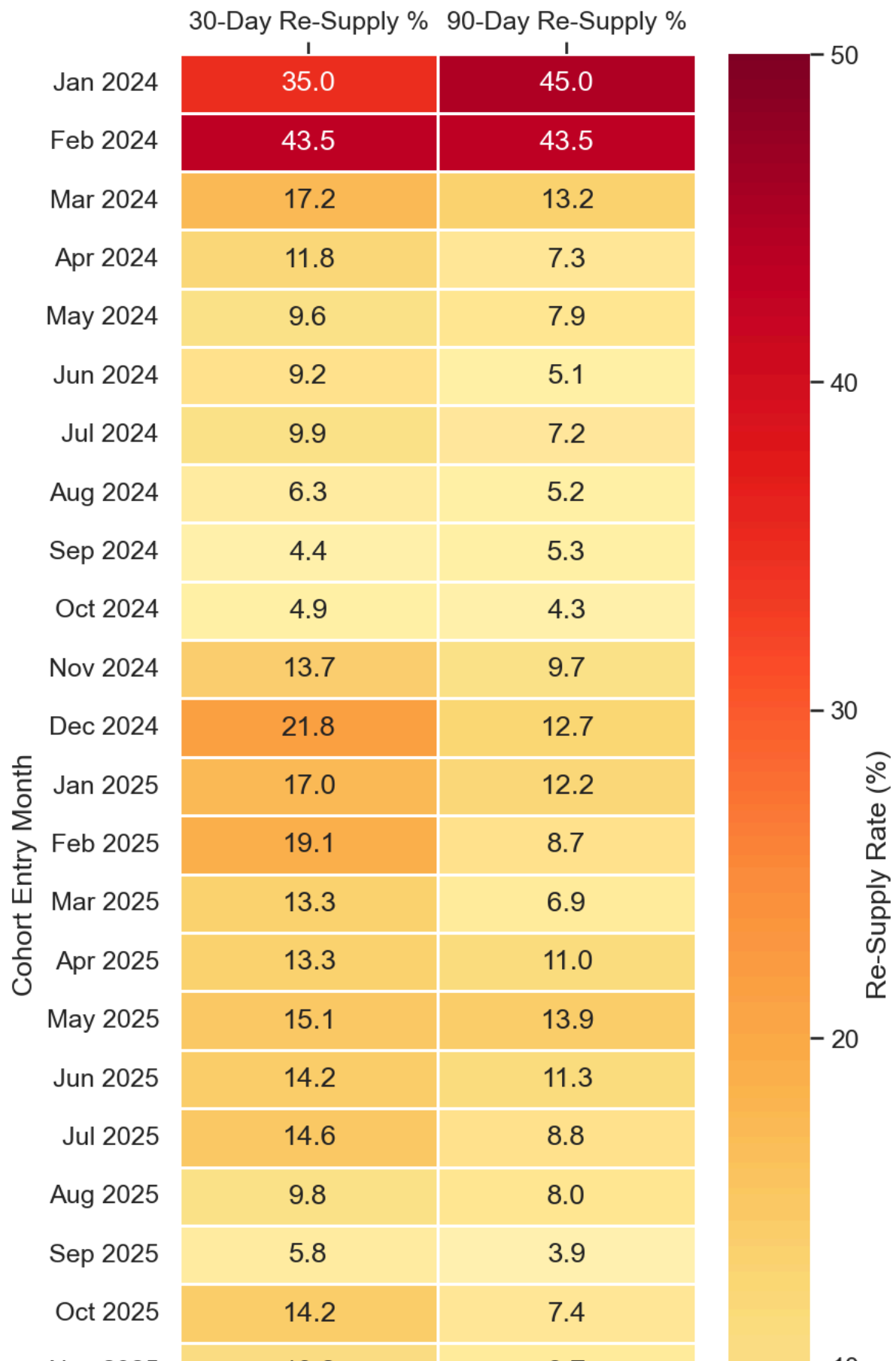
For institutional partners, a live-balance-weighted age near 137 days means the average surviving dollar has been in the protocol for roughly four and a half months, a useful baseline when sizing credit facilities or assessing counterparty exposure. For risk teams, the 92.8-day gap between historical-event and live-balance age is a reminder that aging metrics built on raw supply-event histories will overstate actual capital stickiness whenever withdrawal rates are elevated.

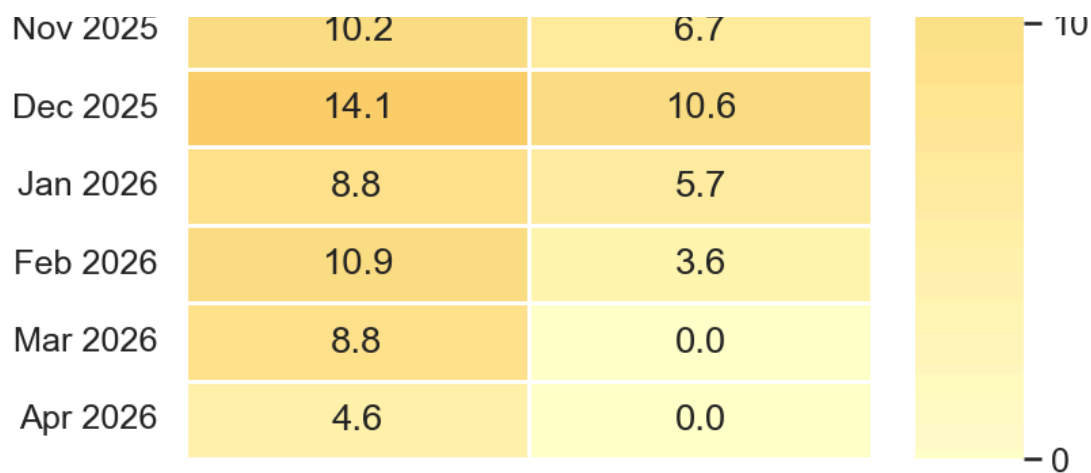
3.5 Cohort Re-Supply Rate (*Activity-Based; Not a Stickiness Proxy*)

Critical interpretation note: This metric is strictly activity-based. A depositor who supplies once and leaves capital untouched for a year is reported as 0% re-supply. Low re-supply rates in mature cohorts are a signal of vault-automation adoption, not disengagement.

Figure 8 — Cohort Re-Supply Rate Heatmap

Morpho Blue — Cohort Re-Supply Rate Heatmap
(Activity-Based Metric: Repeat Supply Events, Not Balance Persistence)





Left column: 30-day re-supply rate. Right column: 90-day re-supply rate. Darker = higher activity.

Results. January–February 2024 cohorts reached 45.0% / 43.5% 90-day re-supply rates (small samples of ~20 wallets each). From March 2024 onward, re-supply rates faded to single digits or low teens and have remained there.

The decline reflects protocol maturation: LP interactions shifted from manual `lending.supply` deposits to automated vault-mediated positions. Smart contracts manage and rebalance positions under the hood; address-level supply transaction frequency decays toward zero while capital remains deployed.

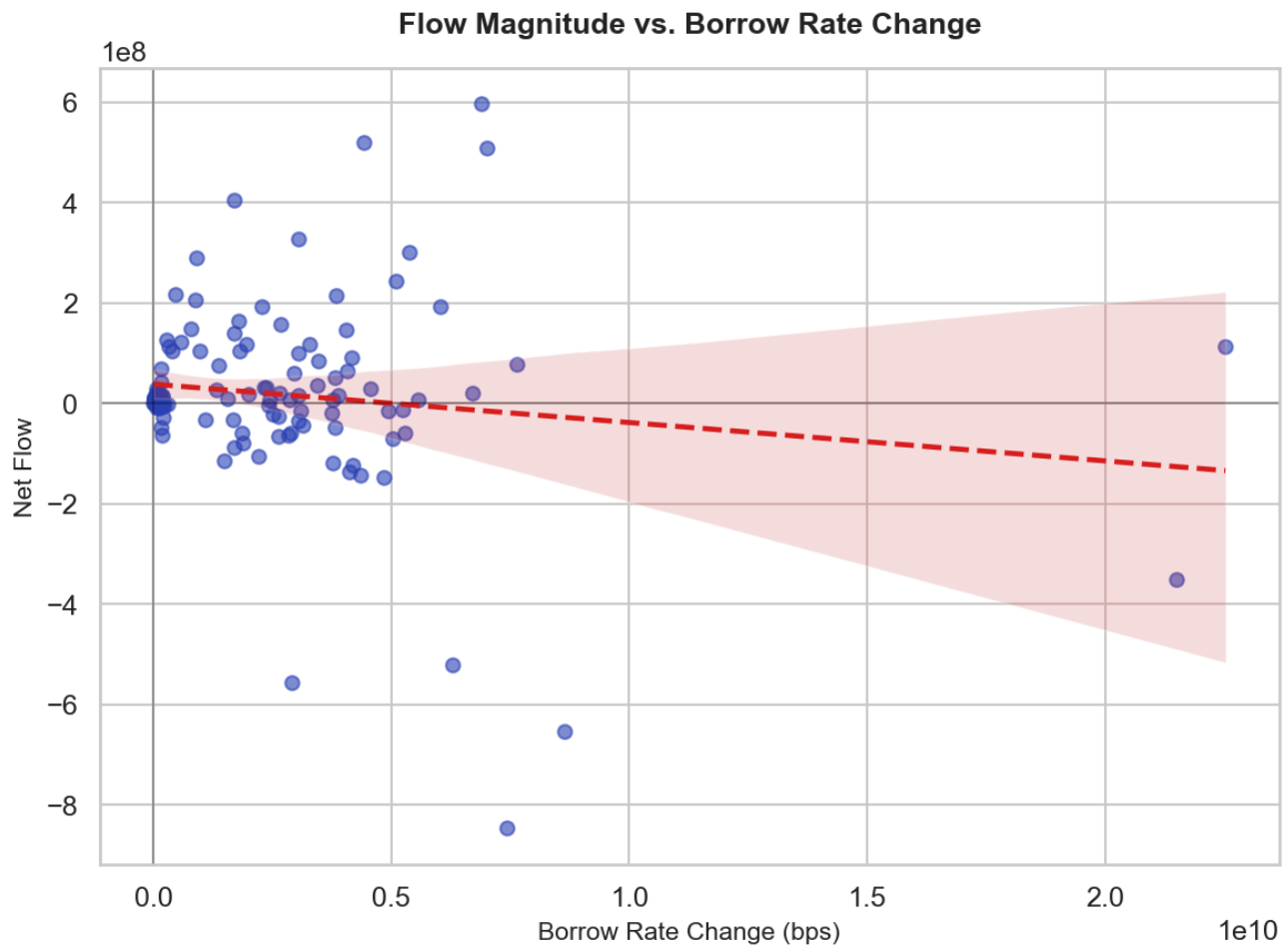
The declining re-supply frequency from early 2024 cohorts onward is a structural artifact of vault adoption, not user disengagement. LPs who moved to vault-mediated positions no longer interact directly with Morpho Blue markets; their capital remains deployed but the address-level supply transaction count goes to zero.

Balance-based survival is the correct instrument for capital stickiness. Re-supply rate measures engagement with the deposit action, not capital persistence.

3.6 Flow Sensitivity to Rates (*Context Metric*)

Figure 9 — Weekly Net USDC Flow vs. Borrow Rate Changes (124 weeks)

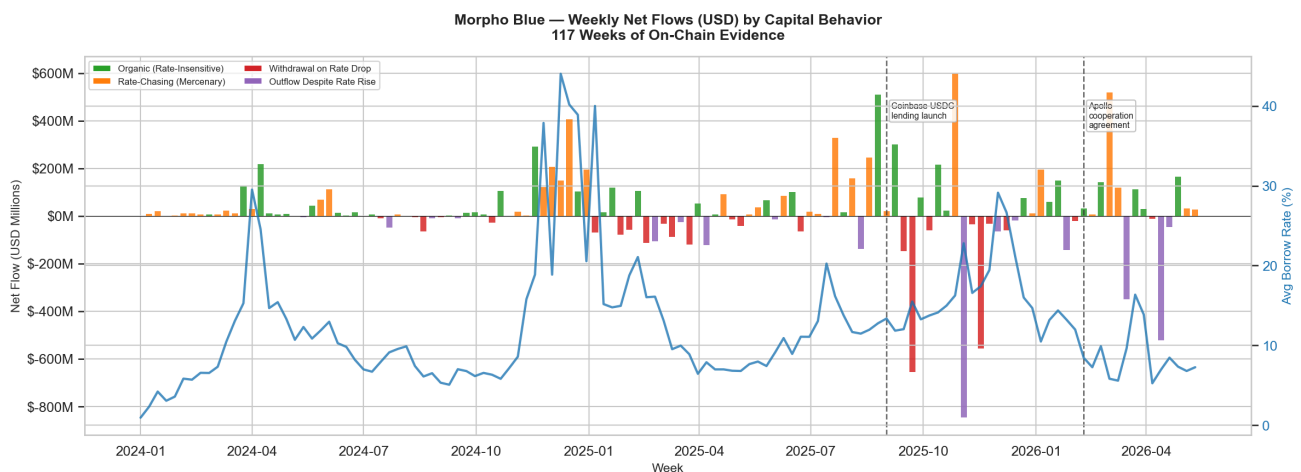
Takeaway: If Morpho’s depositors were purely rate-driven, dots would cluster in the upper-right (rates up → net inflow) and lower-left (rates down → net outflow). Instead, points are present in all four quadrants, but they predominantly cluster in the upper-left, and the regression line is nearly flat, indicating no meaningful linear relationship between borrow rate changes and net flows. The near-flat regression line (red dashes) and the wide, diverging confidence band both confirm $R^2 \approx 0.004$, meaning borrow-rate changes alone explain less than 0.5% of weekly flow variance. The widening confidence band at high rate-change values reflects sparse data at the extremes and should not be read as a signal of stronger sensitivity in tail weeks.



Each dot is one week. The x-axis is the magnitude of that week's borrow rate change (in bps); the y-axis is net USDC capital flowing in or out of Morpho.

Figure 10 — Weekly Net USDC Flows Categorised by Capital Behavior (117 weeks)

Takeaway: Linear regression of weekly net flow magnitude against borrow rate changes yields $R^2 \approx 0.004$, indicating near-zero explanatory power. 32% of weeks show organic, rate-insensitive capital inflows. Factors beyond short-term yield optimization (integrations, security, capital routing efficiency) clearly drive a meaningful share of inflows, but this metric describes weekly flows, not individual depositor behavior, and is correlational only.



Color-coded weekly bars (Organic / Rate-Chasing / Withdrawal on Rate Drop / Outflow Despite Rate Rise) with the average borrow rate overlay (blue line, right axis). The distribution across all four categories in most periods is consistent with the scatter result above.

Table 3.5 — Flow Behavior Distribution (124 weeks)

Behavior	Weeks	Share
Organic (Rate-Insensitive)	40	32.3%
Rate-Chasing (Mercenary)	38	30.6%
Withdrawal on Rate Drop	27	21.8%
Outflow Despite Rate Rise	18	14.5%
Flat / No Change	1	0.8%

Each row is a behavior pattern observed across 124 weeks of data. Organic weeks are ones where capital flowed in despite no yield incentive, a signal of structural adoption. Rate-Chasing weeks are expected mercenary behavior. The roughly equal split between these two categories suggests Morpho's depositor base is not purely yield-driven.

4. Conclusion: Liquidity Duration Synthesis

Morpho Blue's TVL is a tiered capital stack, not a monolithic pool.

Tier	Description	Duration Profile
Foundation	Direct EOA depositors + vault-mediated retail LPs (<code>eoa_underlying</code>)	Median 30d capital survival 37–40%; median hold 80–155 days depending on curator; provides reliable baseline liquidity
Overlay	Professional contract allocators	Rotates within 7–30 days by design; dominates TVL during growth waves
Growth-wave surge	Contract-dominated large cohorts (Sep–Nov 2025)	Half-life ≤ 7 days at aggregate; foundation layer still present but numerically subordinate

For institutional partners and curators building credit products on Morpho, pool-level TVL is an unreliable indicator of usable credit duration. To construct stable, long-dated facilities, partners must:

1. **Segment by address type:** target `eoa_underlying` and EOA bucket metrics, not the `all` rollup
2. **Weight by cohort vintage:** early-vintage institutionally-curated vaults (Re7, Steakhouse, Flagship) carry structurally longer duration than recent growth-wave cohorts
3. **Track the October 2024 regime break:** 90-day `eoa_underlying` survival dropped from ~49% to 11–19% post-Oct 2024; credit facilities priced on pre-transition duration assumptions are miscalibrated

4. **Discount contract-routed capital:** allocator rotation on weekly/monthly cadences is normal operation, not a stress signal; but it means this capital cannot be treated as stable funding for multi-month credit

Metric	Finding	Key Caveat
Balance-Based Survival (primary)	EOA 30d: bimodal 0.03%–35.4%; eoa_underlying median 37–40%; contract near-zero in Sep–Nov 2025	all rollup is dollar-weighted; Oct 2025 shows correlated EOA-contract convergence
Capital Age (historical-event)	229.5d USD-weighted on \$239.2B gross USDC volume	Can age mechanically; not a live-balance measure
Capital Age (live-balance approx.)	~137d; 93 days below historical-event	Cohort-level approximation; 110–160d plausible range
Cohort Re-Supply Rate	Jan–Feb 2024 peak: 45%/43.5% 90d; subsequent cohorts: single digits	Activity-based; passive holders score 0%; reflects vault-automation shift, not churn
Flow Sensitivity to Rates	32.3% of weeks show rate-insensitive inflows; $R^2 \approx 0.004$	Correlational; classifies weeks, not depositors

What the data establishes: Morpho Blue has a structurally distinct, duration-stable supply core (EOA and vault LP) that is systematically obscured by the contract allocator layer in unsegmented analysis. The two layers are not interchangeable for credit duration purposes.

What the data does not establish: A single protocol-wide stickiness number. EOA retention is bimodal and cohort-dependent. The October 2024 regime break in vault LP duration is consistent with institutional scaling but requires allocator-level attribution to confirm. Cross-protocol comparisons are directional only.

The \$239.2B gross cumulative volume figure represents total supply-event activity over time. It should not be compared to current TVL.

Scope: All findings are derived from Morpho Blue Ethereum on-chain supply data. The framework is designed to generalize to other lending venues, but applying it there requires dedicated analysis with harmonized taxonomies.

5. Stakeholder Implications

The aggregate all rollup should not be read as end-user behavior for large 2025 cohorts. Address-type segmentation is the operative lens.

Risk teams: Sub-2% headline survival for Sep–Nov 2025 is a *composition signal* (contract-dominated cohort rotating), not a panic signal. Risk monitoring on the aggregate rollup will produce false-alarm patterns whenever contract dollar share is high. The address-type split distinguishes the two.

Protocol team: Contract-bucket flow is mercenary by design; EOA-bucket flow is materially stickier on the same horizons. The protocol's "real" duration metric is the per-bucket curve (especially `ea_underlying`), not the rollup. Incentive design, risk parameters, and vault listing benefit from separating the two. The Sep–Nov 2025 contract-bucket surge and rapid rotation signals a strong dependency on allocator routing decisions made outside the protocol.

LPs / Allocators: `ea_underlying` and EOA survival is the right benchmark. Across large 2025 cohorts, 30-day EOA survival is highly dispersed (0.03%–35.4%). Segment supply by who controls the wallet and weight cohort-by-cohort rather than averaging. The dispersion is the headline, not the mean.

Vault Curators: The per-cohort EOA survival curve is the most direct on-chain signal of whether an LP base is duration-stable or rotation-prone. Curators optimizing for TVL growth and curators optimizing for stable credit supply face a genuine trade-off: growth waves introduce high-velocity capital that dilutes stickiness metrics.

Researchers: Address-type segmentation should be standard practice for any lending protocol analysis where contract-routed capital is a material share of flows. The methodology (ERC-4626 blindspot resolution + contamination audit + EOA filter) generalizes directly to other protocols, tokens, and chains.

`ea_underlying` **40.2% median 30-day survival** (cohorts $\geq$ \$1M initial USDC, EOA-only per query 7536232, contamination-removed) is the institutionally defensible baseline for Morpho vault LP stickiness within the Morpho-only scope.

6. Future Work

6.1 Cross-Protocol Duration Benchmarking (*highest priority*) Apply identical balance-based survival methodology and address taxonomy to Aave v3 and Compound v3 USDC on Ethereum. Infrastructure (queries 7535541, 7535543) is built. The key test: whether the Oct 2024 regime shift is Morpho-specific or market-wide.

6.2 Allocator-Level Tagging Tag known routers, allocators, and institutional smart vaults to move from the aggregate "contract bucket" to allocator-specific survival and rotation patterns. Primary purpose: test whether the bimodal EOA survival pattern is a curator-composition effect.

6.2a Curator Metadata Enrichment via API The Morpho API directly returns curator metadata (`curators`, `allocators`, `timelocks`, `pendingConfigs`) for any vault. Rather than hand-labeling curator identities (as done in §3.3), future analysis should query the API programmatically to enrich the vault taxonomy with governance config and curator identity. This enables automated cross-vault duration comparisons that account for risk parameters and incentive structures alongside LP stickiness.

6.2b Vault Risk Warnings as a Control Variable The Morpho API's `warnings` field returns structured risk flags (`type` , `level`) per vault. Incorporating API-derived risk conditions into the address-type survival analysis (§3.1) could help isolate whether the bimodal EOA retention pattern reflects curator design, risk stratification, or cohort-entry timing. This is particularly relevant for §5 (Risk teams and Vault Curators) who need to distinguish duration-stable capital from risk-driven exits.

6.5 Borrower-Side and Credit-Stack View Extend framework to the borrower side and stacked positions (Morpho as collateral into other protocols).

6.6 Intra-Protocol Extensions

- Cross-token: USDT, WETH, WBTC (USDC ≈ 62.5% of Morpho deposits)
- Migration-adjusted survival (intra-protocol rotation vs. full withdrawal)
- Per-event live-balance age reconstruction (vs. current cohort-level approximation)
- Re-supply rate segmented by address type

6.7 Cross-Chain and Time-Series

- Replicate for Base, Arbitrum, Hyperliquid deployments
- Rolling survival curve computations to test stationarity of duration metrics quarter-over-quarter

Contact & License

<https://dune.com/0xronman> | https://x.com/ronman_chili | <https://linkedin.com/in/ronith-ry>

Code (notebooks, scripts, utilities) is licensed under the MIT License.

Research report citation: Yalamanchili, Ronith. "Liquidity Duration on Morpho Blue." May 2026.

Appendix: Methodology Detail

Dune Analytics Query Reference

Query	Description
7502787	Balance-based capital survival by cohort x horizon x address type (primary dataset)
7536232	eoas_underlying EOA-Filtered Survival (v8.1) : applies contamination-audit EOA filter to vault share data. Used in this report.
7535425	Contamination Audit : classifies vault share recipients as EOA vs. contract; quantifies 15.28% notional contamination removed in 7536232.
7532526	Vault ERC-4626 Blindspot Resolution (base/pre-filter query)
6871840	Cohort re-supply rate

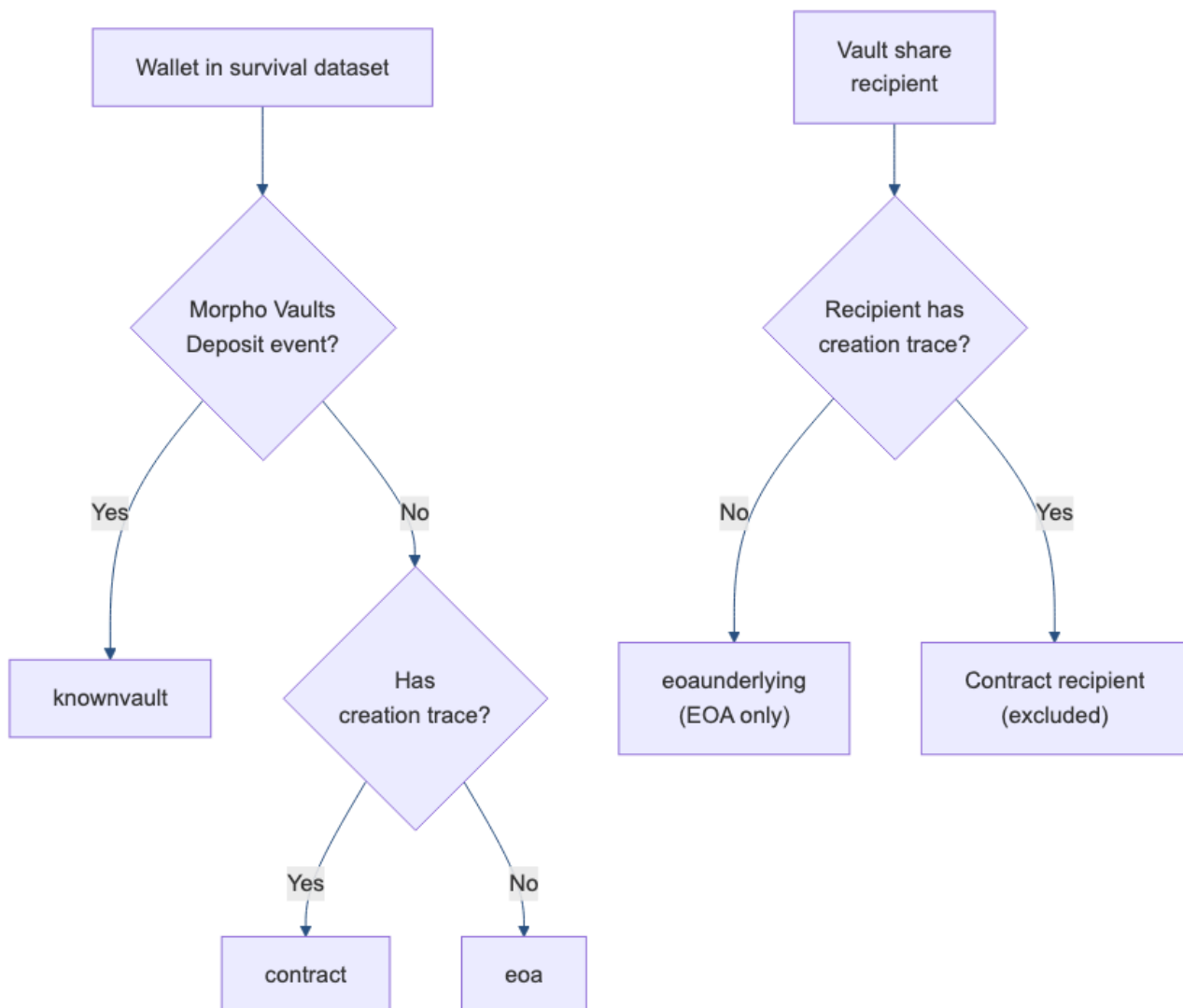
Query	Description
6872058	USD-weighted capital age by loan token
6872073	Weekly net USDC flows vs. borrow rate

Cross-protocol queries (7535541, 7535543: Morpho vs. Aave v3) are built as infrastructure for future work and are not used as primary evidence in this report.

A.1 Address Taxonomy

Each wallet is assigned to one bucket by precedence: `eo_a_underlying > known_vault > contract > eo_a`. The `eo_a_underlying` bucket is sourced by tracking ERC-20 vault share transfers (`erc20_ethereum.evt_Transfer`), filtering out mint/burn events (zero address transfers), then cross-referencing recipients against `ethereum.creation_traces` to exclude deployed contracts.

Figure A.1 — Address Taxonomy Precedence Flow



A.2 Cross-Protocol Calibration (Directional Only)

Over the 18-month mature window (Jan 2024 – Jun 2025), Morpho's EOA-filtered vault-LP retention remains below Aave v3's direct EOA retention, but the gap narrows materially once allocator-routing and vault-abstraction distortions are removed from the comparison. In other words, aggregate Morpho retention metrics understate the stickiness of underlying LP capital because a large share of observable flow is mediated through short-duration routing contracts rather than end-user wallets. This comparison is directional only: Morpho's `eoa_underlying` proxy and Aave's direct EOA deposits are not structurally equivalent, and differences in modular versus monolithic protocol architecture affect how capital movement appears on-chain.

Protocol	30d Survival (capital-wtd)	90d Survival (capital-wtd)
Morpho Blue (<code>eoa_underlying</code> proxy)	38.3%	22.4%
Aave v3 (EOA direct)	58.0%	47.7%
Compound v3 (EOA direct)	53.0%	41.3%

A.3 ERC-4626 Vault Share Pipeline (EOA Filter)

To build the `eoa_underlying` bucket: (1) isolate all MetaMorpho vault ERC-20 `Transfer` events; (2) exclude mint/burn events (zero address); (3) cross-reference every recipient against `ethereum.creation_traces` ; (4) exclude addresses with creation traces (deployed contracts). Remaining addresses are true EOAs representing beneficial vault ownership.

Figure A.2 — Vault Share Pipeline & EOA Filter

